DOST Form 2 (for Basic/Applied Research)

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(1) PROJECT PROFILE

Project Title: Establishing RadioJOVE and SkyNet for Astronomical and Exoplanet Research: Application of White Noise Analysis of Curated Datasets
Project Leader/Sex: Dr. Christopher C. Bernido / Male
Project Duration (number of months): 12 months
Project Start Date: September 2025
Project End Date: August 2026
Implementing Agency: Central Visayan Institute Foundation
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Consultants:
Dr. Reinabelle Reyes, Director, *Space Science Missions Bureau*, *Philippine Space Agency*
Dr. Michael Rutkowski, Associate Professor, *Minnesota State University, USA*
Dr. Xavier Bacalla, Assistant Professor, *University of San Carlos*
Dr. Jingle Magallanes, Professor, *Mindanao State University - Iligan Institute of Technology*

(2) COOPERATING AGENCY/IES (Name/s and Address/es)

(3) SITE(S) OF IMPLEMENTATION

Central Visayan Institute Foundation (CVIF); Research Center for Theoretical Physics (RCTP)

(4) TYPE OF RESEARCH	**(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)**
__x__ Basic _____ Applied	_____ Agriculture, Aquatic and Natural Resources Commodity: ________ _____ Health Priority Topic: ________ __x__ Industry, Energy and Emerging Technology Sector: Space Science/ Astronomy-Astrophysics _____ Disaster Risk Reduction and Climate Change Adaptation __x__ Basic Research Sector: Space Science/ Astronomy-Astrophysics
Sustainable Development Goal (SDG) Addressed	SDG 9: Industry, **Innovation**, and Infrastructure SDG 4: Quality Education SDG 17: Partnerships for the Goals

(6) EXECUTIVE SUMMARY (not to exceed 200 words)

Despite the Philippines' geographical advantage for sky observation, the country lags behind in radio and astronomy observations, with national space science efforts so far concentrated mainly on satellites through the Philippine Space Agency (PhilSA). This project introduces pioneering use of **RadioJOVE** and **SkyNet Robotic Telescope Network** in the Philippines, bridging low-cost instrumentation, global datasets, and local capacity building. The project is organized into four components:

1. **Assembly and Operation of RadioJOVE Telescopes** – Setting up radio telescope systems to monitor solar and planetary radio emissions, including solar flares and solar bursts.
2. **Remote Access to SkyNet Robotic Telescopes –** Applying for and utilizing telescope time to create a pipeline and conduct exoplanet transit observations guided by NASA's Exoplanet Archive data and contribute to the worldwide scientific investigations on exoplanets.
3. **Curation and Analysis of Datasets** – Collecting, organizing, and applying stochastic white noise analysis to local and external datasets to detect patterns and anomalies.
4. **Capacity Building and Sustainability** – Delivering workshops and trainings supported by Learning Activity Sheets (LAS) that serve as instructional tools for educators, students, and researchers.

By combining instrumentation, data science, and training, this project aims to establish a foundation for astronomy research while positioning the Philippines as an emerging contributor to global astronomy efforts.

(7) INTRODUCTION

(7.1) RATIONALE/SIGNIFICANCE (not to exceed 300 words)

Astronomy and astrophysics remain underdeveloped research frontiers in the Philippines despite the country's unique geographical advantage as an equatorial observing site. While neighboring nations have active astronomical facilities, the Philippines has very limited capacity for systematic data collection in planetary, solar, and exoplanetary science. This creates a gap in both regional and global scientific contributions.

This project seeks to pioneer radio astronomy and exoplanet studies in the country by integrating two complementary approaches: (1) RadioJOVE systems for ground-based detection of solar and Jovian radio emissions, and (2) remote access to SkyNet robotic telescopes for optical monitoring of exoplanet transit events. Combining these platforms will enable multi-wavelength observations, offering students and researchers novel opportunities to engage in frontline astrophysics research.

At present, local capacity for astronomy is constrained by the lack of instrumentation, structured data pipelines, and training in advanced analytical methods. By establishing operational expertise in radio observations, data processing pipelines, and stochastic modeling, this project will contribute to national capability-building and scientific self-reliance. Moreover, the integration of local datasets with international archives such as NASA's Exoplanet Archive ensures that Philippine-based efforts are globally relevant.

The long-term significance lies in positioning the Philippines as an active participant in global astronomy networks while fostering a new generation of Filipino researchers in physics and data science. The project will not only produce baseline datasets but also establish methods that can be extended to other astrophysical studies, thus contributing to both national scientific development and the international body of knowledge in astronomy.

(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK

The Philippines is geographically positioned along the equatorial region, an advantageous site for astronomical observations. However, radio astronomy and exoplanet studies remain underexplored locally, with current national space science efforts largely focused on satellites. Establishing observational capabilities with the RadioJOVE system and remote access to SkyNet telescopes provides a cost-effective entry point for Philippine-based research in space science.

RadioJOVE allows detection of radio emissions from the Sun and Jupiter. These signals are important because they reveal solar activity patterns and Jupiter's strong magnetospheric interactions. Yet, no systematic studies have been conducted in the Philippines to document which of these signals can be consistently recorded under local atmospheric and electromagnetic conditions. Addressing this gap is essential in determining the feasibility of long-term radio astronomy research in the country.

Complementing this is optical observation of exoplanet transits through SkyNet telescopes. By combining observational data with established archives such as NASA's Exoplanet Archive and Harvard's DIY Planet Search, exoplanet parameters (transit depth, orbital radius, and period) can be derived. Developing a localized, optimized pipeline for such analysis will contribute to Philippine participation in international exoplanet science.

Central to the project is the curation of datasets for downstream stochastic white noise analysis. White noise stochastic methods are powerful in identifying hidden structures, anomalies, or recurring patterns in time-series astronomical data. These approaches, though widely applied in physics and computational sciences, remain novel in Philippine astronomy research.

Finally, the project integrates a capacity-building dimension. By training faculty and students in telescope operation, data processing, and stochastic analysis, measurable gains in research participation and outreach will be realized. This addresses both the scientific and educational gaps in Philippine astronomy, making the project both pioneering and sustainable.

With this, the specific research questions explored in this project include:

1. What radio signals from the Sun and Jupiter can be detected and recorded in the Philippines using the RadioJOVE system?
2. What exoplanet parameters (e.g., transit depth, period, orbital radius) can be derived using the project-developed optimized data processing pipeline that integrates SkyNet observations alongside NASA's Exoplanet Archive and Harvard's DIY planet search?
3. What quantity and type of datasets on radio astronomy and exoplanets can be compiled and curated into a usable database for white noise stochastic downstream analysis?

4. What measurable patterns or anomalies can be identified in radio astronomy and exoplanet datasets using stochastic white noise analysis?
5. What measurable gains in student or faculty research participation, training, or outreach can be achieved through the implementation of the project?

(7.3) OBJECTIVES

General:

1. To establish and operate the first RadioJOVE-based radio observation station in the Philippines.
2. To develop and validate a data analysis workflow that processes SkyNet telescope data alongside NASA's Exoplanet Archive and Harvard's DIY planet search.
3. To generate, organize, and analyze datasets related to RadioJove radio astronomy and exoplanet observations and apply stochastic white noise analysis for scientific pattern detection.

Specific:

a) To assemble and calibrate a fully functional RadioJOVE antenna and receiver system for detecting solar and Jovian (Jupiter) radio bursts.
b) To record and archive radio signal data from the Sun and/or Jupiter under local conditions using the RadioJOVE system.
c) To obtain and process exoplanet transit datasets from SkyNet observations.
d) To implement a custom Python-based pipeline that enables exoplanet candidate selection, as well as parameter estimations using SkyNet, Harvard's DIY planet search, and NASA data.
e) To curate a structured dataset on astronomical data.
f) To apply stochastic white noise analysis to whichever dataset is applicable.
g) To produce Learning Activity Sheets (LAS) that function as educational toolkits and step-by-step implementation guides for students, educators, and researchers—intended for use in classroom settings, training workshops, and outreach programs, and made accessible for long-term use.

(8) REVIEW OF LITERATURE

The proposed project will build on existing global and local efforts in radio astronomy, exoplanet research, and science education, with particular focus on low-cost observatories, remote telescope networks, and advanced data analysis techniques. While initiatives such as RadioJOVE and SkyNet have already shown success in engaging both researchers and students, the Philippines still lacks a dedicated program that integrates these tools with stochastic white noise analysis and sustained educational outreach. The following studies provide the scientific and educational foundation for this work.

- Globally, radio astronomy and exoplanet science are advancing rapidly, with networks of observatories and space missions enabling discoveries on cosmic phenomena and planetary systems. In the Philippines, astronomy research is steadily growing, supported by academic institutions (Guido, 2019) and the establishment of the Philippine Space Agency (PhilSA). However, the country currently lacks a dedicated radio astronomy observatory (Masong & Pobre, 2017), making this project pioneering in its aim to establish the first such facility. While PhilSA's primary programs remain focused on satellite development and Earth observation (*2024 Annual Report Philippine Space Agency*), complementary initiatives in radio astronomy and exoplanet research are essential to broaden the nation's scientific and technological capabilities.

- The RadioJOVE Project, developed by NASA, provides a low-cost, community-supported radio telescope system that enables the observation of solar and Jovian radio emissions (Higgins, 2014). RadioJOVE stations worldwide have contributed data to study solar bursts, space weather events, and Jupiter's magnetosphere. However, despite its global reach, a review of active RadioJOVE participants, online literature, and the NASA RadioJOVE archive indicates that no dedicated station has yet been established in the Philippines, highlighting the need for this project to pioneer such an initiative locally.

- The SkyNet Robotic Telescope Network, operated by the University of North Carolina, has been instrumental in making optical astronomy accessible to students and researchers worldwide (Schlekat et al., 2025). By integrating SkyNet into the Philippine research ecosystem, this project will provide local learners with access to high-quality astronomical imaging and participation in exoplanet transit monitoring campaigns, bridging the gap between global facilities and local needs. The transit method for detecting exoplanets—where dips in starlight reveal the passage of a planet—has been widely documented in NASA's Exoplanet Archive and applied in citizen-science initiatives such as Harvard's DIY Planet Search (Brennan et al., n.d.). Through NASA's EXOTIC software, which processes light curves from these transit events, students can replicate the same analytical steps used by professional astronomers. Embedding this methodology into classroom-ready research modules will allow Philippine students, such as those at CVIF, not only to visualize stellar light curves but also to practice the validation of exoplanet discoveries in real data.

- White noise stochastic analysis provides a framework for studying systems where random fluctuations exhibit memory effects rather than being purely Markovian. By comparing mean square displacements (MSDs) of datasets with theoretical predictions, deviations from classical diffusion can reveal memory effects. From this, parameters of memory functions can be extracted, showing how past states influence present dynamics (Bernido & Carpio-Bernido, 2014). Applications have extended to biophysical data such as probe diffusion in fibrin gels (Aure et al., 2019) and ecological datasets like coral reef decline (Elnar et al., 2021). This framework is also being applied to astronomical data such as in determining the maximum of solar cycle (Toledo et al., 2024). These studies highlight the versatility of white noise analysis in bridging empirical datasets with theoretical models, offering tools relevant for analyzing astrophysical signals in projects like RadioJOVE and SkyNet.

- Finally, research on astronomy education and outreach emphasizes the importance of engaging learners using real astronomical data and authentic research tools to enhance scientific literacy and foster lasting interest in science (Pompea & Russo, 2020). However, many educational outreach initiatives—often inspired "flash" events like stargazing nights or seasonal workshops—struggle to sustain engagement over the long term. For example, a review of high school astronomy student-research projects over the past two decades found that funding is frequently tied to short-term grants, and many programs collapse into volunteer-run efforts once support expires (Fitzgerald et al., 2014). Similarly, in the Philippines, public astronomy activities have largely been limited to one-off initiatives such as stargazing camps and public lectures organized by the Philippine Astronomical Society (PAS) (Philippine Astronomical Society, 2023). While valuable, these efforts often lack continuity, underscoring the need for more structured and sustainable approaches to astronomy engagement. In contrast, Bernido & Bernido (2020) demonstrated the effectiveness of Learning Activity Sheets (LAS) in sustaining learning (Bernido & Bernido, 2020). Building on this evidence, the project adopts LAS-based astronomy outreach to provide continuity and deeper engagement, outlasting flash events like stargazing nights. By aligning classroom-based LAS with real observational data, the project contributes both to scientific capacity-building and to the UN Sustainable Development Goal 4: Quality Education.

(9) METHODOLOGY

1. Establishment of Radio Astronomy Capability (RadioJOVE)

- **Assembly and Setup**
  - Construct and calibrate RadioJOVE dual-dipole antenna at the rooftop of Central Visayan Institute Foundation.
  - Ensure azimuth/elevation orientation and noise minimization (e.g., rooftop installation, grounding).
- **Data Acquisition**
  - Regular observations of solar bursts and Jovian radio emissions.
  - Recording metadata (date, time, polarization, signal strength, weather conditions).
- **Data Management**
  - Develop standardized logging templates.
  - Store raw data in a centralized repository (local + cloud backup).

2. Optical Observations via SkyNet

- **Credit Purchase**
  - Coordinate with Dr. Michael Rutkowski (Astrophysicist consultant from Minnesota State University) and Dr. Dan Reichart (Founder of the SkyNet Robotic Telescope Network and faculty member at the University of North Carolina) for the purchase of telescope credits.
  - Allocate approximately USD 1,700 (~PHP 100,000), equivalent to around 81,600 SkyNet credits, for project observations. The total observing time will vary depending on the efficiency and availability of the specific robotic telescopes used.
- **Coordination**
  - Access SkyNet UNC robotic telescopes for optical photometry.
  - Align observation requests with exoplanet transit schedules.
- **Observation Protocol**
  - Generate light curves from SkyNet pipeline.
  - Validate transit signals using NASA Exoplanet Archive parameters.
- **Integration**
  - Store SkyNet data alongside RadioJOVE observations for multi-modal dataset.

3. Data Curation and Integration

- **Compilation**
  - Collect local (RadioJOVE, SkyNet) and external (NASA, Exoplanet Archive, Exoplanet Transit Database, AAVSO) datasets.
- **Standardization**
  - Convert datasets to common formats (CSV, FITS where applicable).
  - Annotate with metadata (location, instrument, conditions).
- **Database Development**
  - Build a searchable data repository accessible to the research team.

4. Stochastic White Noise Analysis (Mean Square Displacement Modelling)

- **Rationale**
  - Apply mathematical modelling techniques used in stochastic processes with memory.
- **Process**
  - Compute empirical Mean Square Displacement (MSD) curves from time-series data.
  - Fit empirical MSD against theoretical MSD equations.
  - Derive model parameters (e.g., diffusion constants, correlation times).
- **Application**
  - For RadioJOVE: Identify patterns in solar bursts or Jovian emissions.
  - For Exoplanets: Detect subtle variations in transit timing or depth possibly masked by noise.
- **Validation**
  - Cross-compare derived parameters with known physical/astronomical models.

5. Computer Specifications Required for Data Processing

- **Rationale**
  - The project will handle large astronomical datasets from NASA missions (e.g., TESS and Kepler), which reach hundreds of gigabytes to terabytes in size. Processing these, alongside RadioJOVE and SkyNet data, requires high-performance computing to ensure efficient analysis, visualization, and model fitting.
- **Specifications and Justification**
  - **Processor**: Intel i9 or AMD Ryzen 9 (8 cores or higher) – Provides adequate computational power for parallel data processing and efficient handling of large astronomical datasets (e.g., FITS and light curve files). The high core count and clock speed enable faster execution of data analysis scripts, noise filtering, and model fitting tasks essential for radio and optical data processing.
  - **Memory**: 16–32 GB RAM – Ensures stable performance when loading, visualizing, and analyzing multiple large FITS and CSV files simultaneously. Adequate memory prevents system lag during multi-threaded data processing and enables smooth execution of Python-based analysis tools.
  - **Primary Storage:** 1–2 TB NVMe SSD – Facilitates rapid data access and writing during batch analyses of light curve and radio signal files. The high read/write speed of NVMe drives significantly reduces loading time and improves performance when handling large datasets or running simultaneous data-processing tasks.
  - **Graphics Processing Unit (GPU):** NVIDIA RTX 3050 / 4060 or higher – Supports GPU-accelerated computations for machine learning and white-noise analyses. The GPU's parallel architecture enables faster numerical modeling and signal processing, which enhances efficiency when applying stochastic or time-series algorithms on large datasets.
  - **Connectivity:** WiFi 6, USB 3.0, and Ethernet ports – Required for reliable data logging, RadioJOVE receiver integration, and large dataset downloads.
- **Summary**
  - The above specifications justify the acquisition of a high-performance laptop (approximately ₱160,600) necessary for data-intensive processing, integration of multi-modal astronomical datasets, and advanced computational modeling aligned with the project's scientific objectives.

6. Procurement of Telescopes: Celestron NexStar Evolution 8 and SeeStar S50

- **Rationale**
  - To enable both manual and automated observations for research, workshops, and outreach.
  - To capture high-quality optical data for planets, deep-sky objects, and the Sun, complementing RadioJOVE solar burst measurements.
  - The Evolution 8 provides a large-aperture, high-resolution platform suitable for research-grade imaging.

    - The SeeStar S50 provides automated, participant-friendly imaging, ideal for demonstrations and rapid data collection.
- **Process**
    - Evolution 8: Equipped with ISO-certified solar filter for safe solar imaging; GoTo WiFi control allows precise tracking and integration with cameras.
    - SeeStar S50: Built-in solar filter and automated tracking for easy, app-controlled imaging.
    - Using both telescopes together allows simultaneous participant observation and automated data acquisition.
- **Application**
    - Research: High-resolution imaging of deep-sky and planetary objects with Evolution 8; solar imaging provides optical context for RadioJOVE data; SeeStar S50 allows automated time-series observations.
    - Workshops / Outreach: Participants operate Evolution 8 for hands-on experience and use SeeStar S50 for automated demonstrations, including solar imaging and timelapse sequences.
- **Justification**
    - The Evolution 8 delivers research-grade imaging for deep-sky, planetary, and solar targets, complementing RadioJOVE observations.
    - The SeeStar S50 ensures safe, automated imaging for participant-friendly use.
    - Using both telescopes together maximizes research output, participant engagement, and educational impact, supporting comprehensive astronomical study.

7. Learning Activity Sheets (LAS), Workshops, and Capacity Building

- Development
    - Translate technical concepts (radio waves, exoplanet transits, stochastic modelling, coding, and machine learning) into simplified LAS formats aligned with DepEd K–12 competencies.
    - Create a structured set of LAS that future participants (teachers, students, and community members) can use for astronomy and data science engagement.

- **Infrastructure: Network Attached Storage (NAS)**
    - Establish a centralized NAS to store RadioJOVE data, SkyNet optical observations, NASA mission datasets, and processed results.
    - Use the NAS as a long-term digital repository for all LAS, codes, tutorials, and research outputs, ensuring continuity for future batches conducting astrophysics research.
    - Provide secure, organized access for Science Research Specialists and future researchers, supporting data sharing, reproducibility, and sustainable program growth.

- **Implementation: Workshops and Public Engagement**
    - Conduct hands-on workshops where participants use LAS and project telescopes (NexStar Evolution 8, SeeStar S50, RadioJOVE) for practical learning in optical and radio astronomy.
    - Host the 2nd Jagna Space Encounters, in partnership with the Philippine Space Agency, as a flagship project workshop aimed at strengthening national capacity in astronomy and astrophysics research.
- **Capacity Building for Science Research Specialists (SRS)**
    - Allow the two SRS to attend at least two conferences, workshops, or trainings related to astronomy, data analysis, noise modelling, telescope operation, or science education.
    - Build technical and pedagogical skills that will strengthen the project's research output and enhance their ability to write high-quality LAS for future cohorts.
    - Ensure the team can independently operate instruments, process datasets, and teach emerging astronomy topics beyond the project's duration.
- **Outreach and Sustainability**
    - LAS and archived datasets stored in the NAS become reusable teaching and research resources, enabling long-term astronomy engagement even after the project ends.
    - Strengthened SRS capacity ensures continuous improvement of materials and training in succeeding years.

8. Monitoring and Evaluation

- **Project Tracking**
    - Monthly logs of observations, datasets, and analyses.
- **Workshops**
    - Feedback forms from participants.
- **Research Outputs**

- Number of datasets curated, models fitted, LAS created, students engaged, and papers published.

(10) TECHNOLOGY ROADMAP (if applicable) (use the attached sheet)

(11) EXPECTED OUTPUTS (6Ps)

- Scientific paper focused on radioastronomy and exoplanet analysis.
- Curated Dataset from astronomical observations.
- Trained Science Researchers having the capability to operate RadioJOVE systems and SkyNet protocols, as well as apply white noise stochastic analysis to astronomical datasets.
- Workshops for students and teachers through LAS-based education.

(12) POTENTIAL OUTCOMES

RadioJOVE Observatory Setup – a dual-dipole radio telescope installed at CVIF to record solar and planetary radio emissions, publicly accessible for student immersion and training. Both men and women especially students, are encouraged to conduct data gathering observation using the system.

Scientific Paper – scientific paper with results of the project on radio observations, exoplanet transits, and stochastic white noise analysis will be written.

Learning Activity Sheets on Astronomy and Data Science – simple guides and activity sheets serving as modules for students and teachers, integrating NASA EXOTIC, SkyNet, and RadioJOVE data into classroom use. This will openly available to both men and women especially students.

Public Engagement through Astronomy Outreach – selected findings and educational materials will be presented in school-based seminars, public talks, or online platforms to raise awareness of astrophysics in the Philippines. This will involve three (3) male and two (2) female facilitators.

(13) POTENTIAL IMPACTS (2Is)

- Enhanced knowledge and awareness of students and the community on space science and radio astronomy, inspiring future scientists and innovators.
- Through strengthened capacity in data science and astronomy, local researchers and students gain skills applicable not only in astrophysics but also in industries such as telecommunications, ICT, and disaster preparedness

(14) TARGET BENEFICIARIES

- Undergraduate and senior high school students in physics, engineering, and astronomy-related tracks who will gain hands-on training in space science and data analysis.
- Faculty and researchers in CVIF and collaborating institutions who will develop expertise in radio astronomy, robotic telescope operation, and exoplanet science.
- Philippine Space Science initiatives (e.g., PhilSA programs) by contributing local capacity in astronomy and space observation.
- Broader scientific community through publications and shared datasets on solar, planetary, and exoplanet observations.
- Eventually, the local community in Jagna, Bohol, as the project will enhance STEM education and inspire future space science careers.

(15) SUSTAINABILITY PLAN

- The project will establish baseline infrastructure (RadioJOVE setup and SkyNet access protocols) that will remain functional for continued use in teaching and student research beyond the funding period.
- CVIF faculty and trained students will serve as the next-generation implementers, ensuring knowledge transfer and continuity.
- Integration into the CVIF science curriculum and student research projects will sustain regular use of the equipment and data.
- External partnerships with PhilSA, SkyNet, and academic collaborators will be maintained to ensure continued access to telescope resources and updated datasets.
- Future funding will be pursued through PhilSA research grants, DOST-GIA renewals, and possible inclusion in institutional budgets for science education and outreach.

(16) GENDER AND DEVELOPMENT (GAD) SCORE (refer to the attached GAD checklist)

16.33 (see attached GAD checklist for scoring details). Given the outreach and education nature of the project, participation of women faculty, researchers, and students is highly encouraged. This translates to women empowerment, equipping them with the necessary skills for future scientific endeavors related to astronomy and astrophysics.

(17) LIMITATIONS OF THE PROJECT

- The project depends on fair weather for ground-based telescope observations; poor weather may limit data collection.
- RadioJOVE equipment has limited sensitivity compared to large professional arrays; data resolution will be modest.
- Internet connectivity is needed for SkyNet telescope access; interruptions may delay observations.
- Research scope is limited to solar, planetary radio waves, and exoplanet transit data analysis, not full-spectrum astrophysics.
- The project is focused in Jagna, Bohol, and may not yet reach broader Philippine schools until replication and scaling are supported.

(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN (List possible risks and assumptions in attaining target outputs or objectives.)

Risks and Mitigation:

1. **Equipment malfunction or antenna misalignment**
   - Mitigation: Provide training for staff and students; maintain spare parts and technical support from RadioJOVE community.
2. **Power outages or unstable electricity supply**
   - Mitigation: Use UPS or backup generator to protect equipment and ensure continuity of observations.
3. **Internet connectivity interruptions**
   - Mitigation: Schedule telescope sessions ahead; store and re-download missed data when stable connection returns.
4. **Weather disturbances (typhoons, heavy rains, cloud cover)**
   - Mitigation: Flexible observation scheduling; prioritize data analysis tasks during unobservable days.
5. **Limited technical expertise among first-time users**
   - Mitigation: Training workshops, mentorship from consultants, and step-by-step manuals for operation and data processing.
6. **Sustainability beyond project period**
   - Mitigation: Integrate activities into CVIF curriculum and seek follow-up funding from DOST/PhilSA or institutional support.

Assumptions:

- Collaborating agencies (PhilSA, SkyNet, partner school) will remain supportive throughout the project.
- Students and faculty will remain engaged and available for training and research.
- The global data networks (NASA archives, SkyNet access) will remain open and accessible during the project.

(19) LITERATURE CITED

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(20) PERSONNEL REQUIREMENT

Position	Percent Time Devoted to the Project	Responsibilities
Science Research Specialist I (Licensed Electrical Engineer - Entry Level)	Full Time	• Operate and maintain RadioJOVE and SkyNet tools • Collect and organize observation data • Prepare basic analyses and visual outputs • Assist in reports and technical writing • Help in trainings and outreach
Science Research Specialist II (Licensed Electrical Engineer - Experienced, with Prior Research Background)	Full Time	• Lead data processing and workflow development • Supervise SRS I and Project Assistant • Support in methodology and data interpretation using stochastic analysis • Write technical reports
Project Assistant II (with NTC Radio License)	Full Time	• Handle clerical and administrative tasks • Assist in logistics and equipment needs • Support financial and expense reports • Help organize trainings and outreach

(21) BUDGET BY IMPLEMENTING AGENCY

IMPLEMENTING AGENCY	PS	MOOE	EO	Total
DOST-GIA 7	**1,476,012.00**	**407,800.00**	**448,600.00**	**2,332,412.00**
CVIF	**187,200.00**	**160,000.00**	**130,000.00**	**477,200.00**
Total	**1,663,212.00**	**567,800.00**	**578,600.00**	**2,809,612.00**

TOTAL

(22) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: _____ (number)

Title of the Project	Funding Agency	Involvement in the Project
N/A		

(23) OTHER SUPPORTING DOCUMENTS (Please refer to page 2 for the additional necessary documents.)

1. Project Line-Item Budget (Form 4)
2. Project Work Plan (Form 5)
3. GAD Checklist

I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled is within the prescribed number as stipulated in the DOST-GIA Guidelines. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

	SUBMITTED BY (Project Leader)	ENDORSED BY (Head of the Agency)
Signature	Christopher C. Bernido	Christopher C. Bernido
Printed Name	CHRISTOPHER C. BERNIDO, PhD	CHRISTOPHER C. BERNIDO, PhD
Designation/Title	RCTP Director	CVIF PRESIDENT
Date	September 1, 2025	September 1, 2025

Note: See guidelines/definitions at the back

DOST Form 2 (for Basic/Applied Research)
DETAILED R & D PROJECT PROPOSAL

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the detailed R&D proposal for the component project together with the detailed proposal of the whole Program, project workplan, line-item budget (LIB), 1-page curriculum vitae of the Project Leader, and Certificate of Incorporation or DTI Registration (if applicable) and other applicable supporting documents required under item II.23 below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

II. Operational Definition of Terms:
1. Title- the identification of the Program and the component projects.

Project- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

Project Leader- refers to a project's principal researcher/implementer.

Project Duration- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

Implementing Agency- the primary organization involved in the execution of a program/project which can be a public or private entity

2. Cooperating Agency/ies- refers to the agency/ies that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

3. Site/s of Implementation- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

4. Type of Research- indicates whether the project is basic or applied.
Basic research- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.
Applied research- is an investigation undertaken in order to utilize data/information gathered from fundamental/basic researches or to acquire new knowledge directed primarily towards a specific practical aim or objective with direct benefit to society.

5. R&D Priority Area and Program- based on the Harmonized National R&D Agenda 2017-2022, indicates which R&D agenda the project can be categorized in: Agriculture, Aquaculture and Natural Resources; Health; Industry, Energy, and Emerging Technology; Disaster Risk Reduction and Climate Change Adaptation; and Basic Research. Indicate also the specific Commodity/Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure.

Sustainable Development Goal (SDG) Addressed- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

6. Executive Summary- briefly discusses what the whole proposal is about

7. Introduction- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

7.1. Rationale- brief analysis of the problems identified related to the project
Significance- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

7.2. Scientific Basis- other scientific findings, conclusions or assumptions used as justification for the research
Theoretical Framework- the structure that summarizes concepts and theories that serve as basis for the data analysis and interpretation of the research data.

7.3. Objectives- statements of the general and specific purposes to address the problem areas of the project.

8. Review of Literature- refers to the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific/technical merit; (c) results of related research conducted by the same Project Leader, if any; (d) Prior Art Search, and; (e) other relevant materials.

9. Methodology- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

10. Technology Roadmap (if applicable)**-** a visual document that communicates the plan for technology. It is a flexible planning technique to support strategic and long-range planning by matching short- and long-term goals to specific technology solutions.

11. Expected Outputs (6Ps)- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

Publication- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.

Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.

Product- invention with a potential for commercialization.

People Service- people or groups of people, who receive technical knowledge and training.

Place and Partnership- linkage forged because of the study.

Policy- science-based policy crafted and adopted by the government or academe as a result of the study.

12. Potential Outcomes- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

13. Potential Impacts

Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.

Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

14. Target Beneficiaries- refers to groups/persons who will be positively affected by the conduct of the project.

15. Sustainability plan- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

16. Gender and Development (GAD) Score- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, "Women in Development and Nation Building Act," interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

17. Limitations of the Project- refer to restrictions or constraints in the conduct of the project.

18. Risk- refers to an uncertain event or condition that its occurrence has a negative effect on the project.

Assumption- refers to an event or circumstance that its occurrence will lead to the success of the project.

19. Literature Cited- an alphabetical list of reference materials (books, journals and others) reviewed. Use standard system for citation.

20. Personnel Requirement- details on the position of personnel to be involved in the project, percent time devoted to the project, and responsibilities.

21. Budget By Implementing Agency- personnel services (PS), maintenance and other operating expenses (MOOE), and equipment outlay (EO) requirement of the project by implementing agency for Year 1 and for the whole duration of the project. Please refer to the DOST-GIA Guidelines for the details (Section IX.B of DOST Administrative Order (A.O.) 011, s. 2020).

a. **PS**- total requirement for wages, salaries, honoraria, additional hire and other personnel benefits.

b. **MOOE**- total requirement for supplies and materials, travel expenses, communication, and other services.

c. **EO**- total requirement for facilities and equipment needed by the Program.

22. Other Ongoing Projects Being Handled By the Project Leader- list of ongoing projects being handled by the Project Leader funded by the DOST-GIA Program and other sources, and the accompanying responsibilities relevant to the project.

23. Other supporting documents required- as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program:

a. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]

b. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]

c. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]

d. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]

e. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);

f. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and

g. For the private non-profit/non-government/people's organizations and startups:

i. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority (CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;

ii. Co-signers Statement (if applicable);

iii. Copy of latest Income Tax Return;

iv. Mayor's permit where the business is located;

v. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);

vi. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;

vii. Disclosure of other related business, if any;

viii. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;

ix. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;

h. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and

i. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

III. Criteria for Evaluation:

A. **Criteria for Evaluating Proposals**

Criterion	Definition
Relevance or Significance	Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development
Technical / Scientific Merit	Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner
Budget Appropriateness	The proposed budget is commensurate to the proposed work plan and deliverables.
Competence of Proponent	Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget.

B. **Governing Council / Board and EXECOM's Evaluation Criteria**

Criteria	Indicators	Raw Score
A. Soundness of Proposal (20%)	R&D addresses relevant sectoral need (applicable to pressing concern)	5
	Solution provided is most effective (compared to other proposed solutions)	5
	Proposed budget is reasonable (project is not expensive vis-a-vis output)	5
	Work plan is doable in a given timeframe	5
B. Suitability of Output (30%)	R&D output is cost-effective (cost is competitive in relation to new or existing products or process)	5
	Has identified partners to adopt the technology (with letter of support from the head of the company)	5
	Output can be commercialized (through an existing manufacturer, spin-off or start-up company)	5
	R&D utilization is timely (output should not be overtaken by other solutions)	5
C. Significance of Outcome (30%)	Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI)	5
	Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity	5
	Environment: enhanced environmental health standards, no adverse effect to the environment	5
	Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project)	5
D. Competence of Proponent (20%)	Proponent's expertise aligned with the proposal	5
	Collaboration with relevant agencies and/or industry partners	5
	Thorough understanding of the proposal's deliverables	5
	DOST has good experience with the proponent	5

C. **Additional Criteria on Gender and Development (GAD)**